

2015-2016 – Econ590
Topics in Macroeconomics

Liquidations and Deficient Demands :
Reconciling Hayek's and Keynes' views of
recessions

Paul Beaudry & Franck Portier
franck.portier@TSE-fr.eu

Vancouver School of Economics

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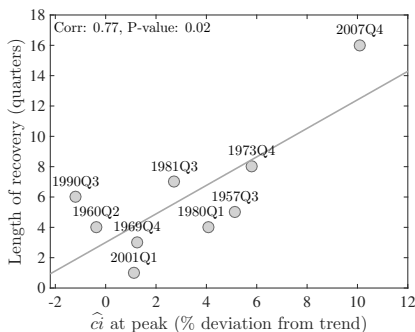
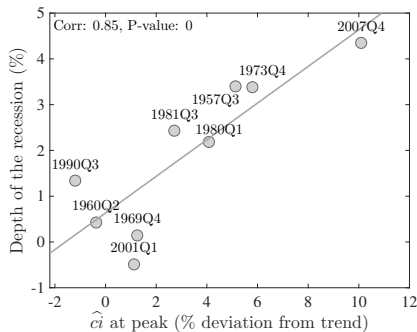
0. Introduction

Recessions

- ▶ Recessions often come after periods of rapid accumulation of assets (productive capital, houses, durable goods)

0. Introduction

Depth of Recession and Length of Recovery vs. Cumulated Total Investment



Note : Horizontal axis is capital over-accumulation, measured as cumulated per capita investment over previous 10 years, detrended using a cubic trend. Vertical axis is either depth of recession, measured as percentage fall in real per capita GDP from peak to subsequent trough, or length of recovery, measured as the number of quarters it takes for real per capita GDP to reach its previous peak. Dates correspond to peak quarters.

0. Introduction

The Relation Between Depth of Recession or Length of Recovery with Cumulated Investment for US Postwar Recessions

$$x_n = \beta_0 + \beta_1 \hat{ci}_n + \varepsilon_n, \quad (1)$$

$$x_n = \beta_0 + \beta_1 \hat{ci}_n + \beta_2 \Delta tfp_n + \varepsilon_n, \quad (2)$$

- ▶ n indexes recessions,
- ▶ x is either the depth of the recession or the length of the recovery,
- ▶ Δtfp_n is the percentage change in TFP from peak to trough
- ▶ $\hat{ci}$ represents deviations from trend of accumulated investment for either total, durable, non-residential or residential capital.

0. Introduction

The Relation Between Depth of Recession or Length of Recovery with Cumulated Investment for US Postwar Recessions

Equation	Depth of recession							
	Tot. Capital		Resid. Capital		N-Res. Capital		Dur. Goods	
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
constant	3.01 (0.06)	3.22 (0.09)	4.41 (0.01)	3.94 (0.04)	3.50 (0.13)	3.71 (0.14)	3.08 (0.09)	3.19 (0.13)
$\hat{c}i_t$	0.94 (0.02)	0.97 (0.03)	0.48 (0.02)	0.48 (0.03)	0.82 (0.15)	0.96 (0.17)	0.74 (0.03)	0.75 (0.05)
Δtfp	- -	-0.19 (0.76)	- -	0.30 (0.63)	- -	-0.42 (0.65)	- -	-0.10 (0.89)

Note : Numbers in parentheses are p-values.

0. Introduction

The Relation Between Depth of Recession or Length of Recovery with Cumulated Investment for US Postwar Recessions

Equation	Length of Recovery							
	Tot. Capital		Resid. Capital		N-Res. Capital		Dur. Goods	
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
constant	0.63 (0.19)	0.88 (0.10)	1.33 (0.02)	1.34 (0.08)	0.63 (0.38)	0.84 (0.21)	0.47 (0.25)	0.70 (0.11)
$\hat{ci}_t$	0.40 (0.00)	0.43 (0.00)	0.17 (0.03)	0.17 (0.05)	0.42 (0.04)	0.56 (0.02)	0.36 (0.00)	0.38 (0.00)
Δtfp	- (0.25)	-0.22 (0.25)	- (0.98)	-0.01 (0.98)	- (0.14)	-0.42 (0.14)	- (0.19)	-0.20 (0.19)

Note : Numbers in parentheses are p-values.

0. Introduction

Recessions

- ▶ Recessions often come after periods of rapid accumulation of assets (productive capital, houses, durable goods)
- ▶ Two opposite views of economic policy in those recessions
 - × HAYEK
 - × KEYNES

0. Introduction

The Liquidationist View (FRIEDRICH HAYEK)

- ▶ Recessions are needed to cleanse the economy.
- ▶ Gvt spendings only delays necessary adjustment
- ▶ *“Any revival which is merely due to artificial stimulus leaves part of the work of depressions undone”*, J. SCHUMPETER, The Economics of the Recovery Program, 1934

0. Introduction

The Aggregate Demand View (JOHN MAYNARD KEYNES)

- ▶ Recessions are periods of insufficient demand
- ▶ Activist fiscal policy is needed
- ▶ *“If the Treasury were to fill old bottles with bank-notes, bury them at suitable depths in disused coal-mines which are then filled up to the surface with town rubbish, and leave it to private enterprise on well-tried principles of laissez-faire to dig the notes up again (the right to do so being obtained, of course, by tendering for leases of the note-bearing territory), there need be no more unemployment and, with the help of repercussions, the real income of the community, and its capital wealth, would probably become a good deal greater than it actually is”, J.M. KEYNES, The General Theory of Employment, Interest and Money, 1936*

0. Introduction

Two opposite views

The Liquidationist View (FRIEDRICH HAYEK)

- ▶ Recessions are needed to cleanse the economy.
- ▶ Gvt spendings, aggregate demand management only delays necessary adjustment

The Aggregate Demand View (JOHN MAYNARD KEYNES)

- ▶ Recessions are periods of insufficient demand
- ▶ Activist fiscal policy is needed

0. Introduction

This Lecture

- ▶ We show that the two views are not mutually exclusive
- ▶ “Over-” (“mal-”) accumulation of physical assets creates the need for *liquidation* $\rightsquigarrow$ recession
- ▶ This *liquidation* will cause the economy to function particularly inefficiently.
- ▶ Some stimulative policies may remain desirable even if they postpone a recovery.

0. Introduction

Main Ingredients

- ▶ Environment with decentralized markets & flexible prices .
- ▶ Two imperfections :
 - × Labor market matching friction in the spirit of Diamond-Mortensen-Pissarides $\rightsquigarrow$ unemployment risk
 - × Adverse selection in the insurance market : unemployment risk is not (fully) insurable.

0. Introduction

What we do not do

- ▶ We do not propose a specific theory of why the economy might find itself with a (too) large stock of houses, durables and/or capital goods.
 - × Noisy news
 - × Lax monetary policy
 - × Exuberance
 - × Perfect foresights limit cycle

0. Introduction

What I will present

- ▶ Static version of model (except for a few slides).
- ▶ Version where “capital” is indeed “durable goods” or “houses” (simpler)

0. Introduction

Roadmap

1. Model setup
2. Equilibrium
3. Interesting Properties of the Static Equilibrium
4. Extensions / Dynamics / Policy Trade-offs

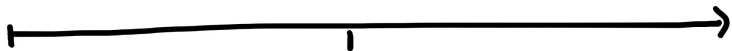
0. Introduction

Roadmap

1. Model setup
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4. Extensions / Dynamics / Policy Trade-offs

1. Model setup

FIGURE 1 – Overview : timeline

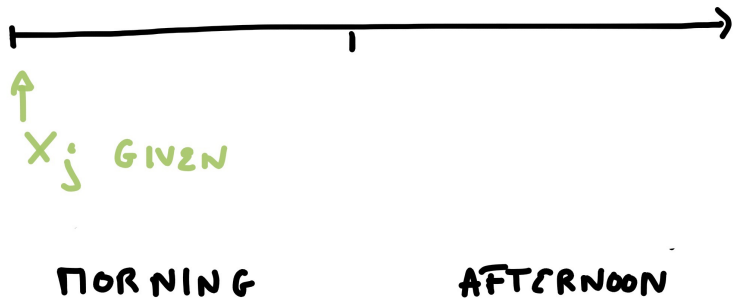


MORNING

AFTERNOON

1. Model setup

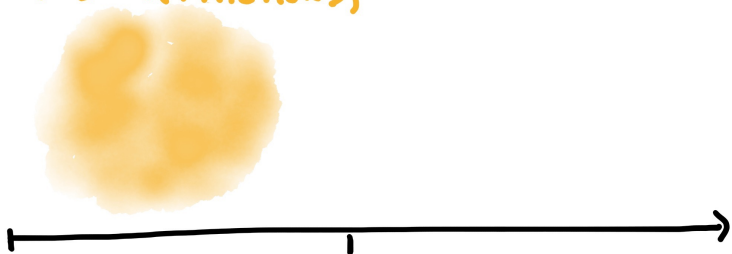
FIGURE 2 – Overview : Initial goods



1. Model setup

FIGURE 3 – Overview : markets

LABOR (FRICTIONS)

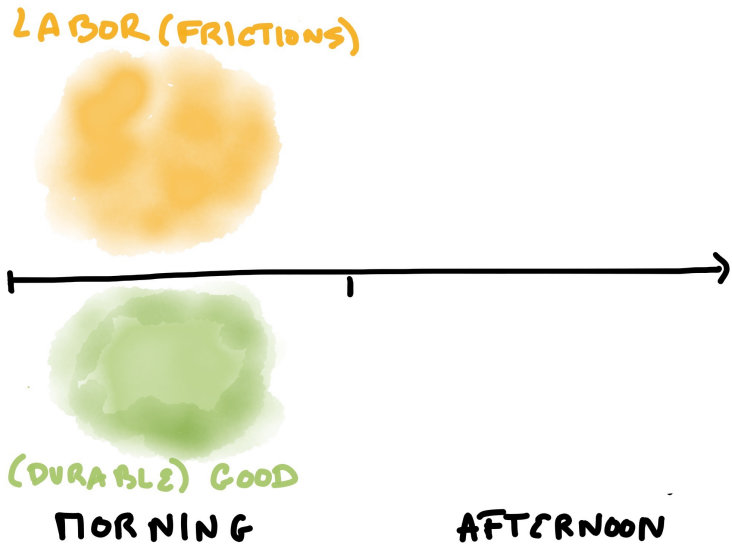


MORNING

AFTERNOON

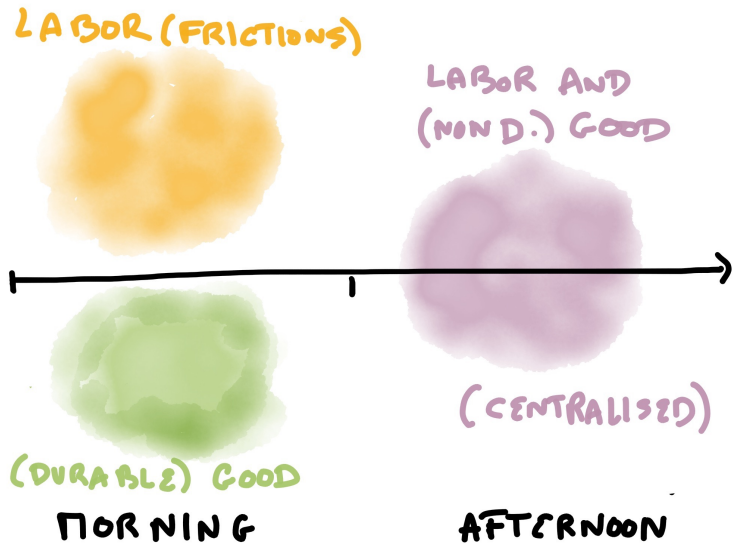
1. Model setup

FIGURE 4 – Overview : markets



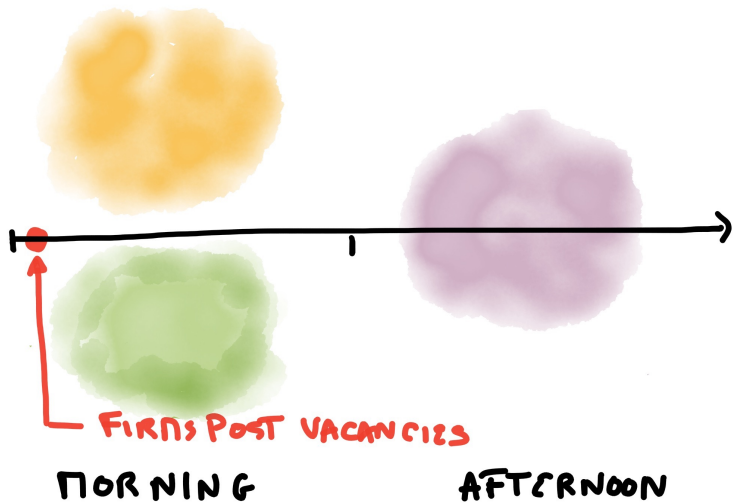
1. Model setup

FIGURE 5 – Overview : markets



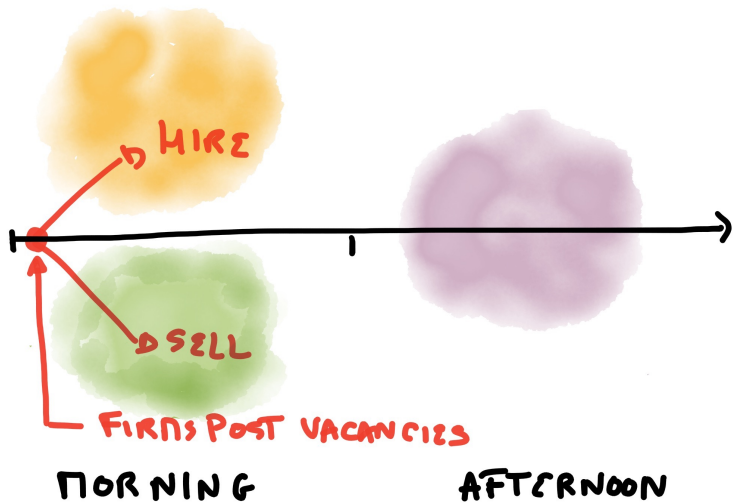
1. Model setup

FIGURE 6 – Overview : firms



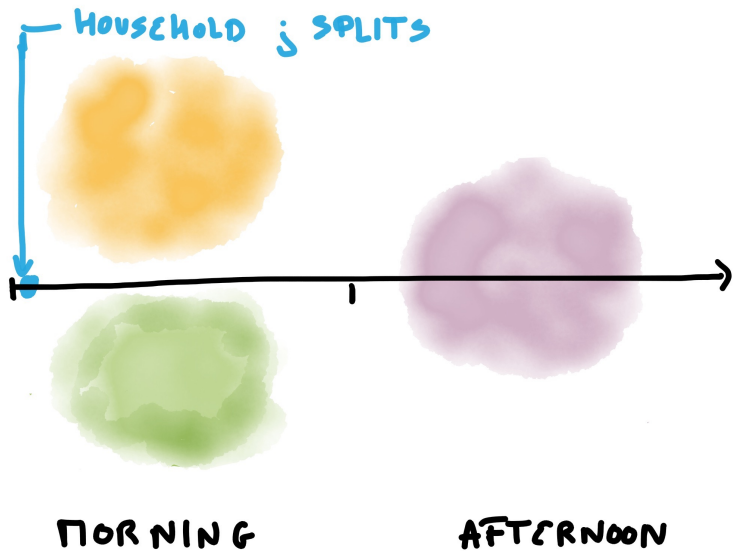
1. Model setup

FIGURE 7 – Overview : firms



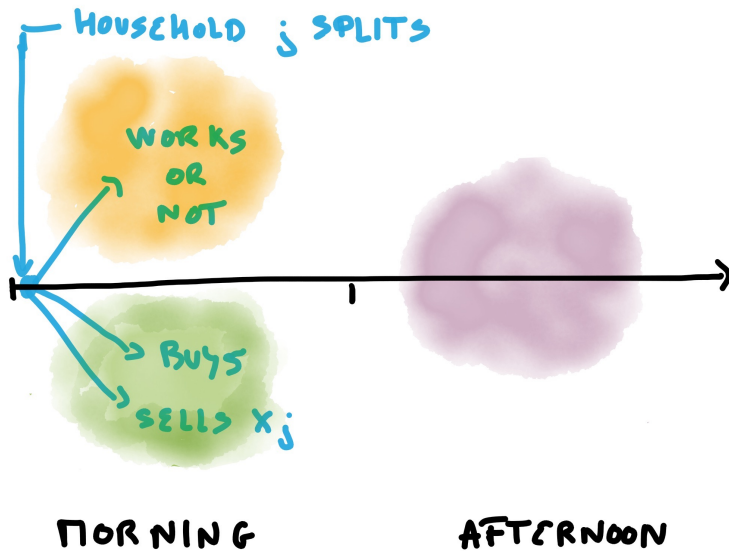
1. Model setup

FIGURE 8 – Overview : households



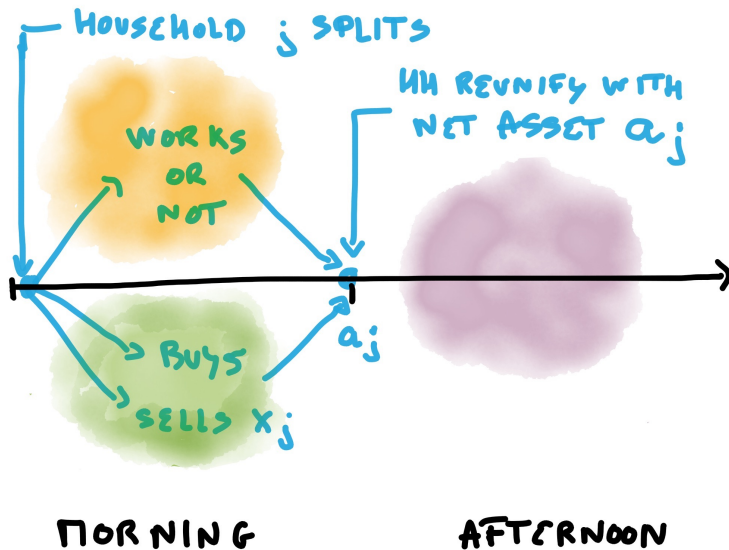
1. Model setup

FIGURE 9 – Overview : households



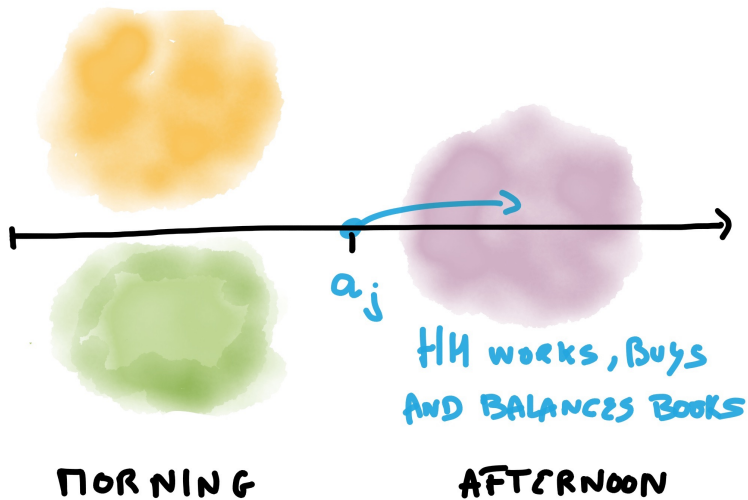
1. Model setup

FIGURE 10 – Overview : households



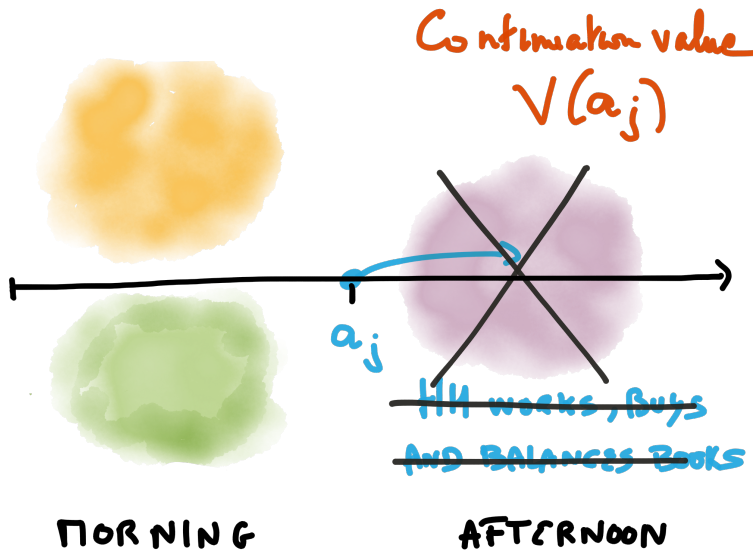
1. Model setup

FIGURE 11 – Overview : households



1. Model setup

FIGURE 12 – Overview : households



1. Model setup

Checklist

- ▶ X_j : exogenous amount of good that is already in household j hands
- ▶ Mass L of households always looking for jobs
- ▶ Afternoon is centralized, all the action is in the morning
- ▶ Frictions on the morning labor market
- ▶ Unemployment risk that is not insured
- ▶ No coordination between firms, shoppers and workers
- ▶ Shoppers and workers credit/debit a bank account that they will clear in the afternoon.
- ▶ Morning good is referred to as “durable” as it will be in the dynamic extension
- ▶ Afternoon good is non durable and serves as the numéraire.

1. Model setup

Preferences



$$U(\underbrace{X_j + e_j}_{c_j}) - \nu(\ell_j) + V(\underbrace{-pe_j + \mathcal{I}_j w \ell_j}_{a_j}).$$

- ▶ Initial endowment of X_j units of durable good.
- ▶ Continuation value $V(a_j)$ given (*in this talk*)
- ▶ $\mathcal{I}_j = \begin{cases} 1 & \text{if employed} \\ 0 & \text{if unemployed} \end{cases}$

1. Model setup

Firms

- ▶ Vacancy posting cost Φ .
- ▶ Decreasing-returns-to-scale production function $F(\ell)$.
- ▶ Net production of a firm hiring ℓ hours of labor from one worker is $F(\ell) - \Phi$.

1. Model setup

Matching

- ▶ N = number firms who decide to search for workers.
- ▶ $M(N, L)$ = number of matches (CRS).
- ▶ “Competitive” match surplus split $\rightsquigarrow$ *within-a-match* hours demand :

$$F'(\ell) = \frac{w}{p}$$

1. Model setup

Normalization and Assumption

- ▶ Normalization : $L = 1$
- ▶ Symmetry : $X_j = X$

1. Model setup

Household morning decisions

- ▶ Worker problem :

$$\max_{\ell_j} -\nu(\ell_j) + \underbrace{V(-pe_j + \mathcal{I}_j w \ell_j)}_{a_j}$$

- ▶ Shopper problem :

$$\max_{e_j} U(X + e_j) + \mu V(w \ell_j - p e_j) + (1 - \mu) V(-p e_j)$$

$\mu \equiv M(N, L)/L$ job finding probability.

1. Model setup

Deriving the value function $V(a)$

- ▶ Not here...
- ▶ $V(a)$ is strictly concave, with the key property that $V'(a_1) > V'(a_2)$ if $a_1 < 0 < a_2$

FIGURE 13 – The Value Function $V(a)$

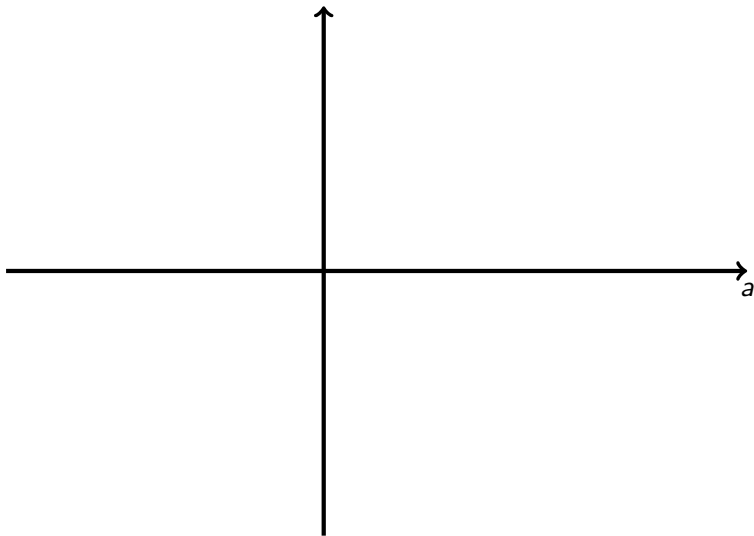


FIGURE 13 – The Value Function $V(a)$

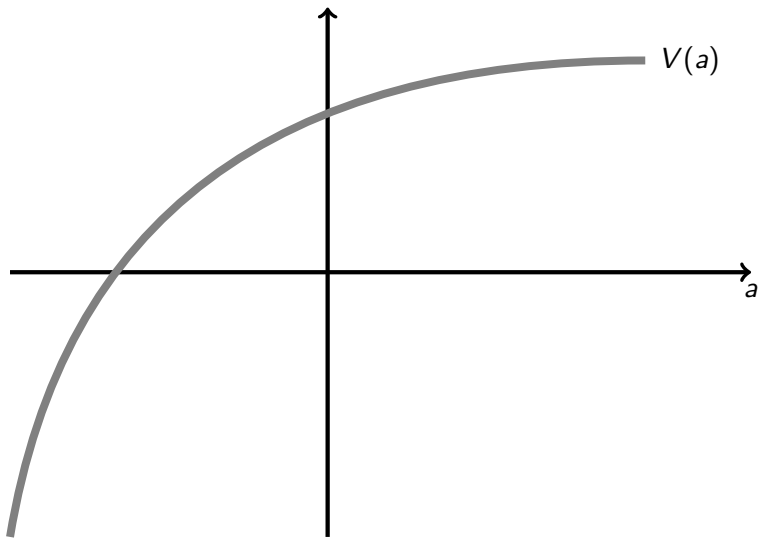


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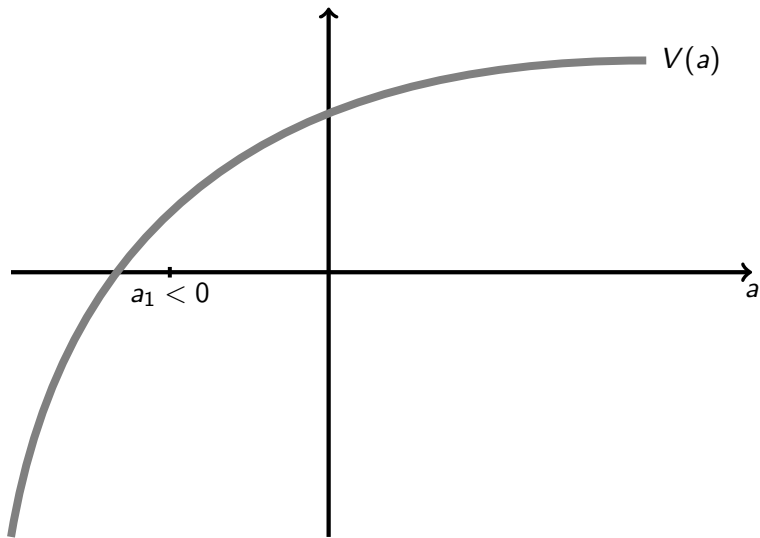


FIGURE 13 – The Value Function $V(a)$

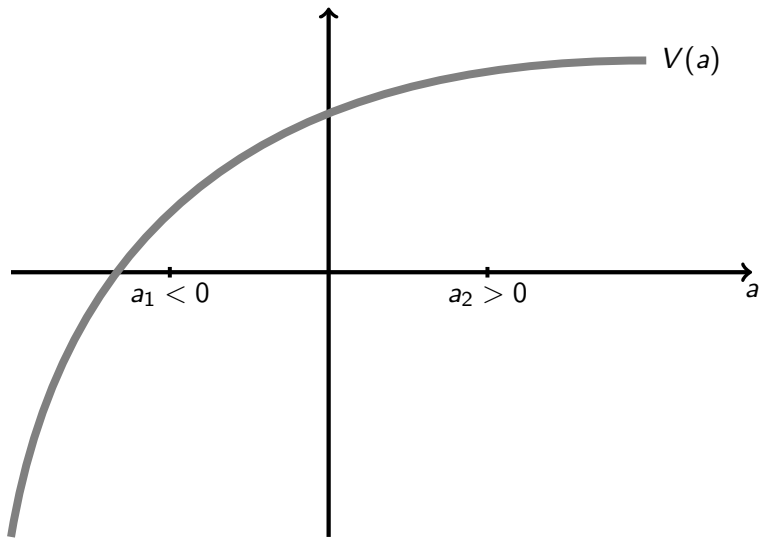


FIGURE 13 – The Value Function $V(a)$

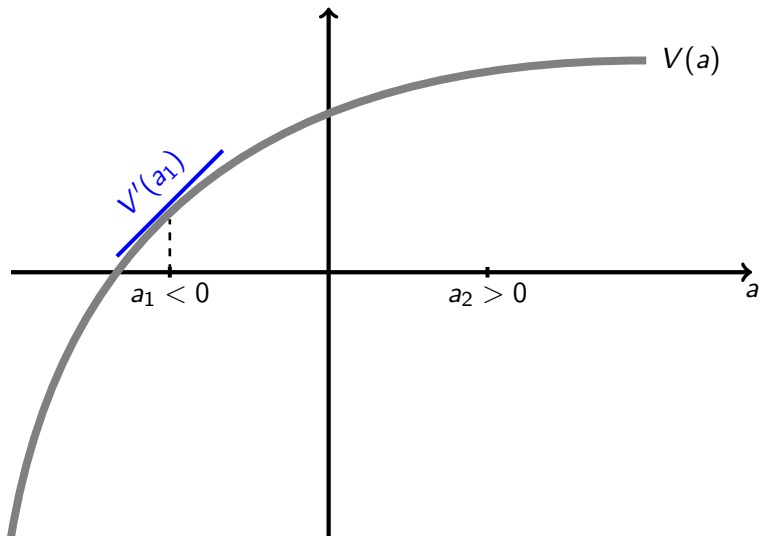
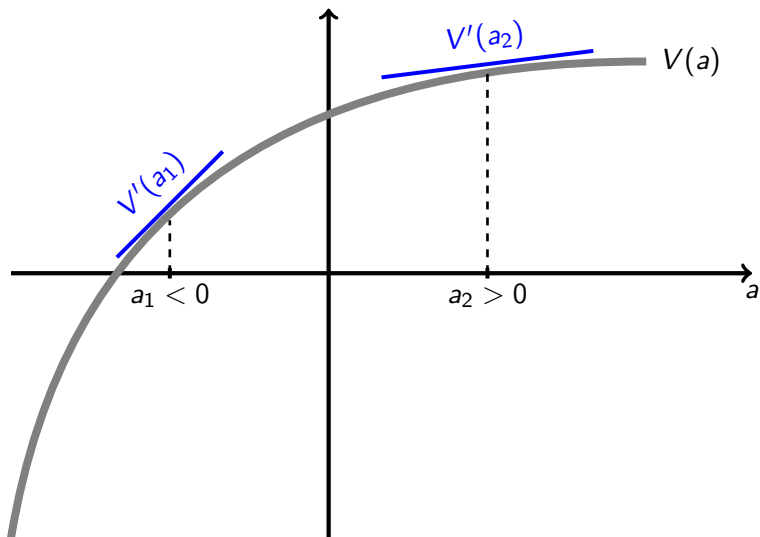


FIGURE 13 – The Value Function $V(a)$



0. Introduction

Roadmap

1. Model setup
2. Equilibrium
3. Interesting Properties of the Static Equilibrium
4. Extensions / Dynamics / Policy Trade-offs

2. Equilibrium

- ▶ Afternoon : $V(a_j)$
- ▶ Morning : markets clear and agents optimize

2. Equilibrium

First sub-period

- ▶ The equilibrium is given by the following equations

▶

$$\begin{aligned}\frac{1}{p}U'(c) &= \frac{M(N, L)}{L}V'(w\ell - p(c - X)) \\ &\quad + \left[1 - \frac{M(N, L)}{L}\right]V'(-p(c - X))\end{aligned}$$

▶

$$\nu'(\ell) = V'(w\ell - p(c - X))w$$

▶

$$pF'(\ell) = w$$

▶

$$\frac{M(N, L)}{N}[pF(\ell) - w\ell] = p\Phi$$

▶

$$M(N, L)F(\ell) = L(c - X) + N\Phi$$

2. Equilibrium

A labor market wedge

$$\frac{\nu'(\ell)}{U'(c)} \underbrace{\left\{ 1 + (1 - \mu) \left[\frac{V'(-p(c - X))}{V'(w\ell - p(c - X))} - 1 \right] \right\}}_{1 + \text{ labor wedge}} = F'(\ell)$$

0. Introduction

Roadmap

1. Model setup
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3. Interesting Properties of the Static Equilibrium
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3. Interesting Properties of the Static Equilibrium

Goal and parametric restrictions

- ▶ Our main goal now is to explore the effects of changes in X on equilibrium outcomes.
- ▶ Why and when an increase in X can actually lead to a reduction in consumption and/or welfare?
- ▶ Can liquidation periods be socially painful?
- ▶ In this talk I restrict the analysis to
 - × $M(N, L) = \min\{N, L\}$

FIGURE 14 – The Matching Function $M(N, L)$

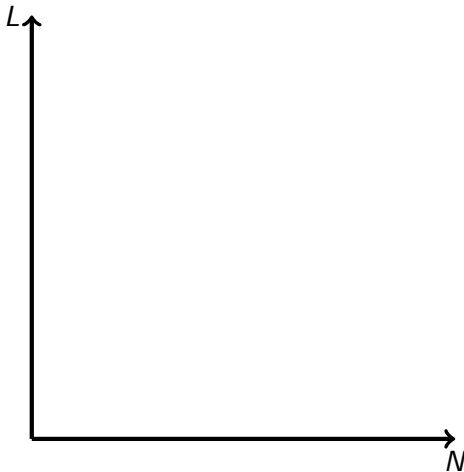


FIGURE 14 – The Matching Function $M(N, L)$

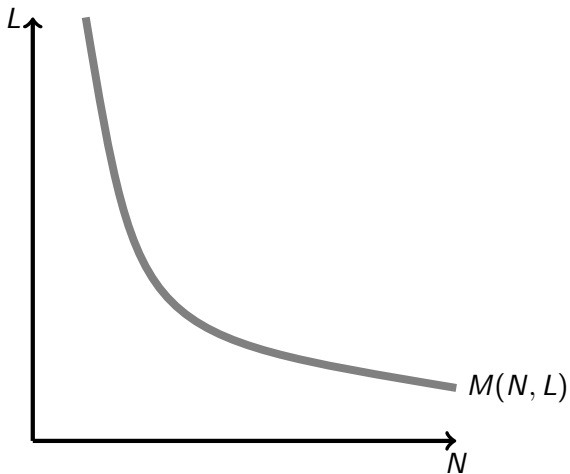


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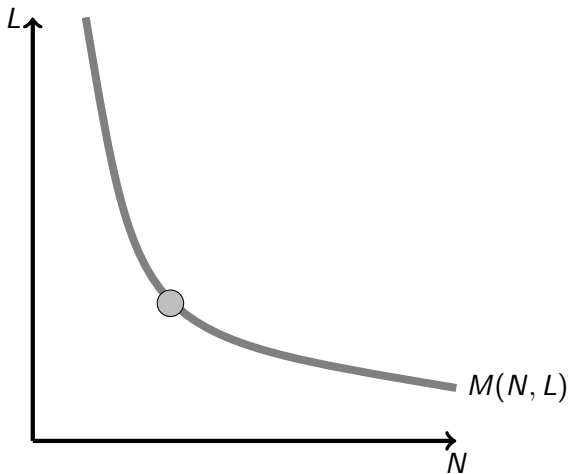
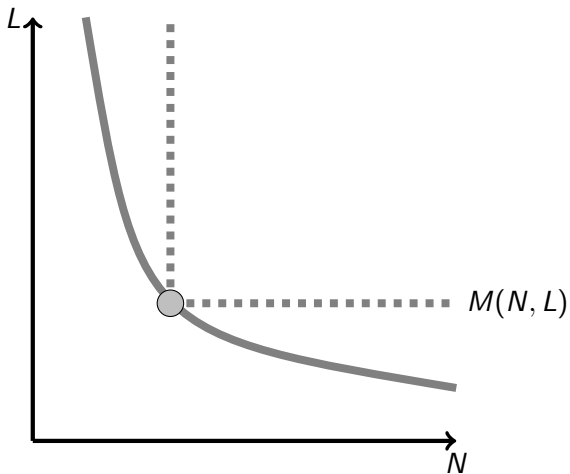


FIGURE 14 – The Matching Function $M(N, L)$



3. Interesting Properties of the Static Equilibrium

Goal and parametric restrictions

- ▶ Our main goal now is to explore the effects of changes in X on equilibrium outcomes.
- ▶ Why and when an increase in X can actually lead to a reduction in consumption and/or welfare?
- ▶ Can liquidation periods be socially painful?
- ▶ In this talk I restrict the analysis to
 - × $M(N, L) = \min\{N, L\}$
 - × $V(a) = \begin{cases} (1 + \tau) \cdot v \cdot a & \text{if } a < 0 \\ v \cdot a & \text{if } a \geq 0 \end{cases}$

FIGURE 15 – The Value Function $V(a)$

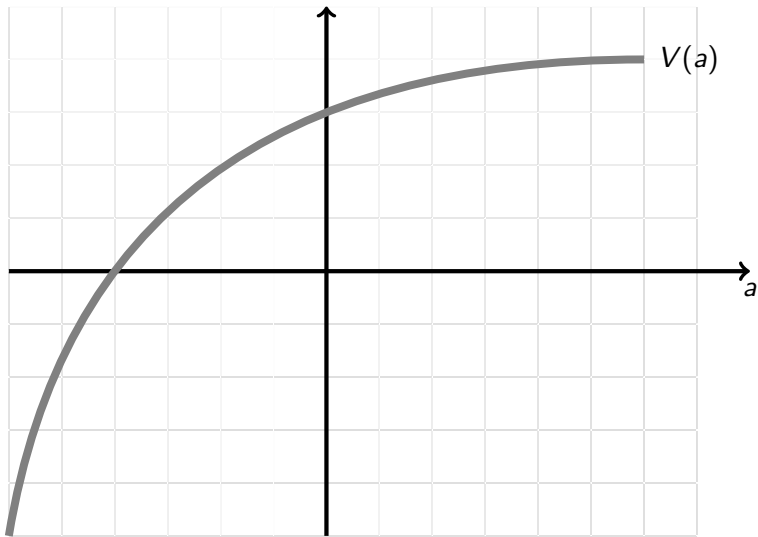


FIGURE 15 – The Value Function $V(a)$

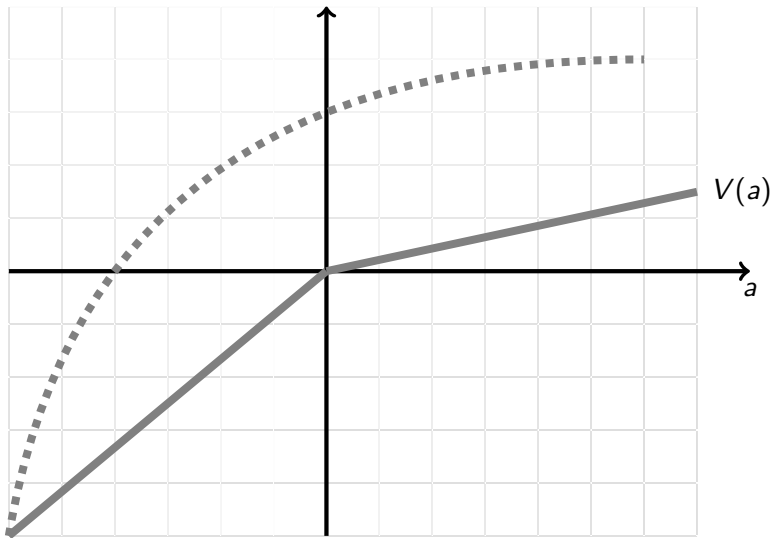


FIGURE 15 – The Value Function $V(a)$

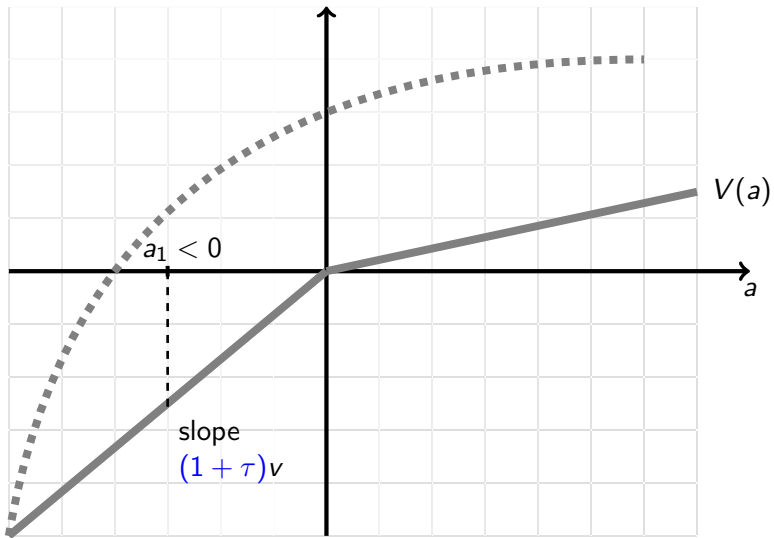
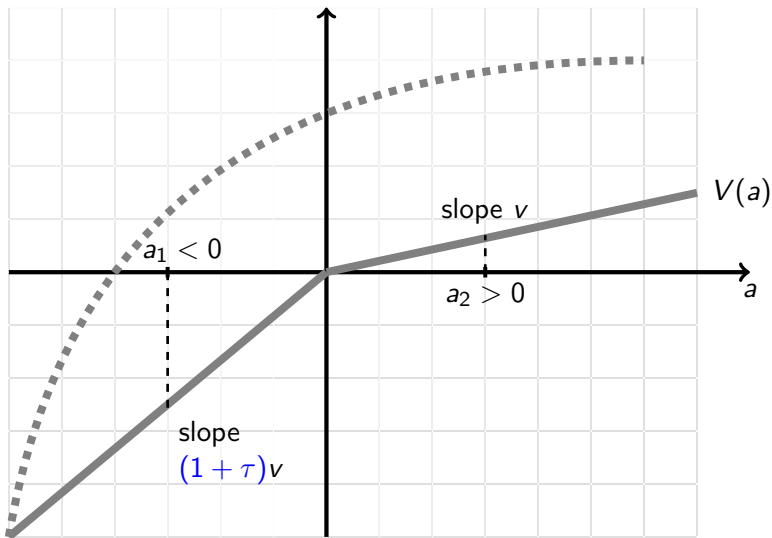


FIGURE 15 – The Value Function $V(a)$



3. Interesting Properties of the Static Equilibrium

With piecewise linear V

$$\frac{\nu'(\ell)}{U'(c)} \left\{ 1 + (1 - \mu) \underbrace{\left[\frac{V'(-p(c - X))}{V'(w\ell - p(c - X))} - 1 \right]} \right\} = F'(\ell)$$

3. Interesting Properties of the Static Equilibrium

With piecewise linear V

$$\frac{v'(\ell)}{U'(c)} \left\{ 1 + (1 - \mu) \underbrace{\left[\frac{V'(-p(c - X))}{V'(w\ell - p(c - X))} - 1 \right]}_{\tau} \right\} = F'(\ell)$$

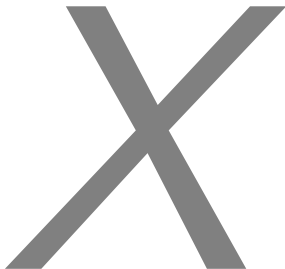
3. Interesting Properties of the Static Equilibrium

With piecewise linear V

$$\frac{v'(\ell)}{U'(c)} \left\{ 1 + \underbrace{(1 - \mu)}_{\mu(X)} \underbrace{\left[\frac{V'(-p(c-X))}{V'(w\ell - p(c-X))} - 1 \right]}_{\tau} \right\} = F'(\ell)$$

3. Interesting Properties of the Static Equilibrium

Two key parameters (1)



3. Interesting Properties of the Static Equilibrium

Two key parameters (2)

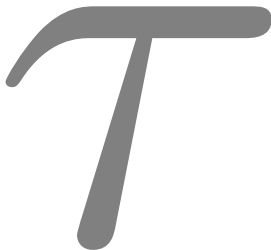


FIGURE 16 – Proposition 1 : Existence and Uniqueness

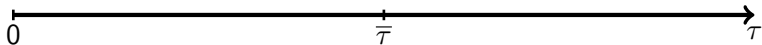


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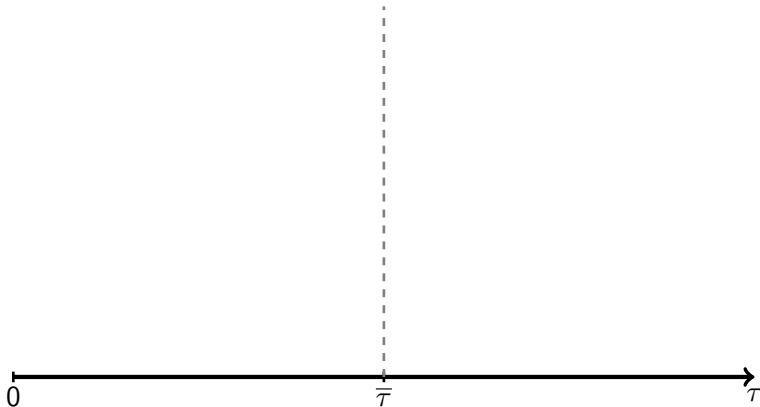


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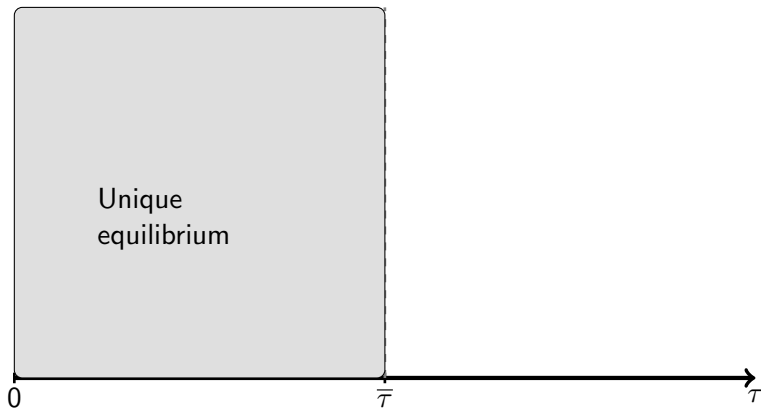


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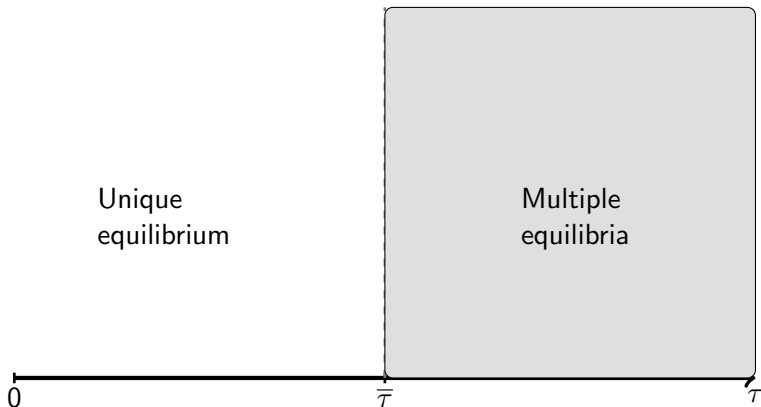


FIGURE 17 – Proposition 2 : The three regimes

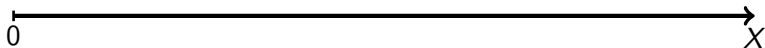


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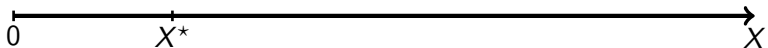


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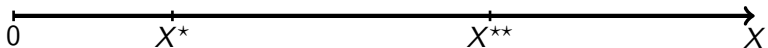


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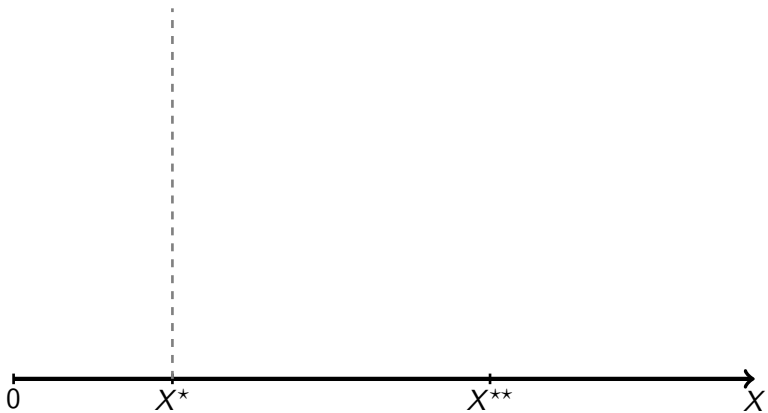


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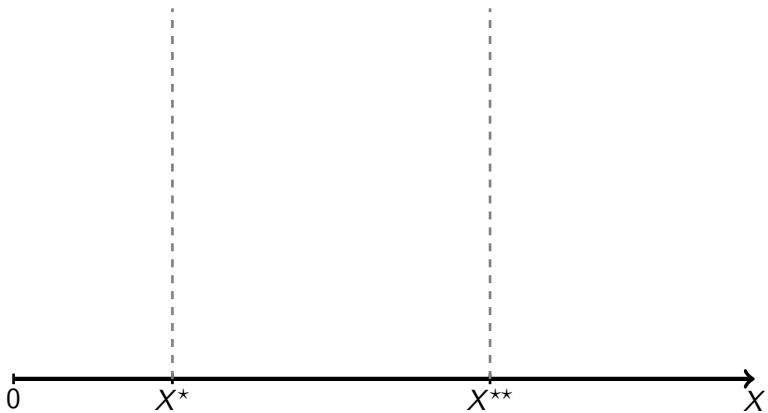


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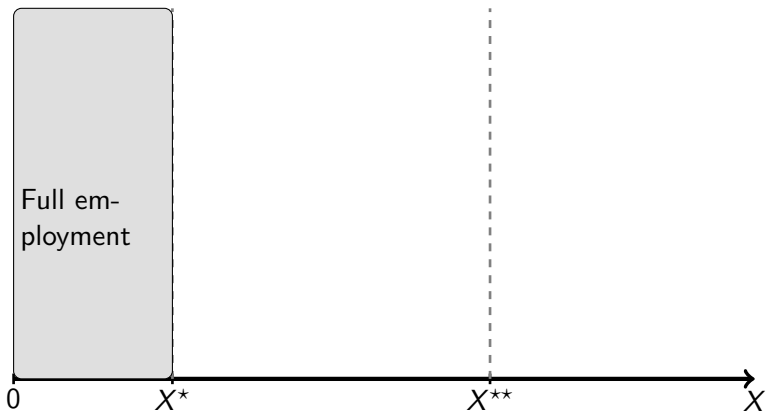


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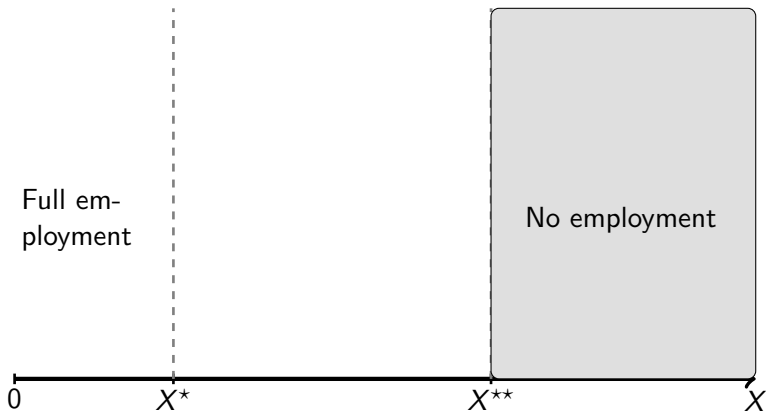
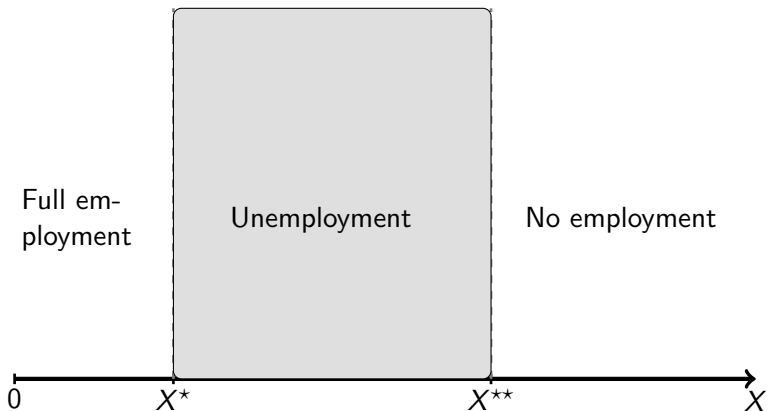


FIGURE 17 – Proposition 2 : The three regimes



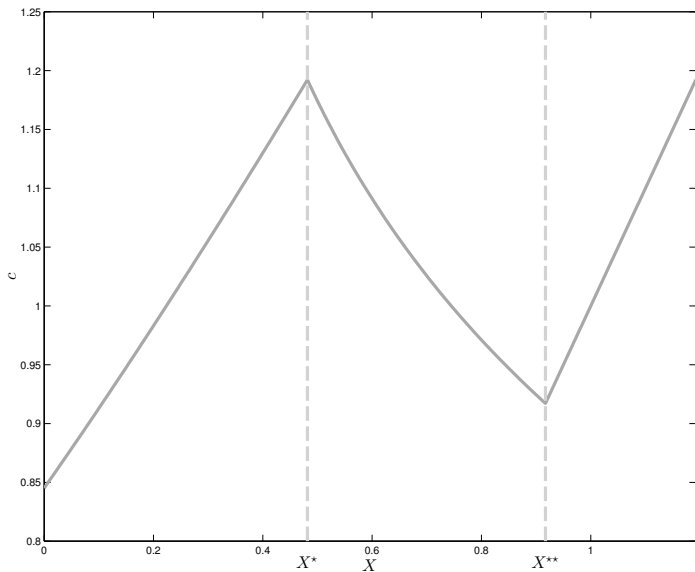
3. Interesting Properties of the Static Equilibrium

Consumption as a function of X

- ▶ How does vary equilibrium consumption when X increases ?
- ▶ In the full employment regime :
 - × Marginal utility of spendings decrease with $X \rightsquigarrow$ less production
 - × But less than proportional to the increase in X
 - × Overall, c increases with X
- ▶ In the no employment regime :
 - × $c = X$
 - × c increases one to one with X
- ▶ In the unemployment regime
 - × “Multiplier > 1 ”
 - × Spendings decrease more than one to one with X
 - × Therefore c *decreases* with X

3. Interesting Properties of the Static Equilibrium

FIGURE 18 – Proposition 3, Consumption as function of X .



3. Interesting Properties of the Static Equilibrium

Is there deficient demand in the unemployment regime?

3. Interesting Properties of the Static Equilibrium

Is there deficient demand in the unemployment regime?

Definition : *Deficient demand* is a situation where

- ▶ increased demand by one agent would favor increased demand by other agents,
- ▶ a feasible coordinated increased in demand by all agents would leave everyone better off.

3. Interesting Properties of the Static Equilibrium

Is there deficient demand in the unemployment regime?

Definition : *Deficient demand* is a situation where

- ▶ increased demand by one agent would favor increased demand by other agents,
- ▶ a feasible coordinated increased in demand by all agents would leave everyone better off.

Proposition 1

When the economy is in the unemployment regime ($X^ < X < X^{**}$), there is deficient demand.*

3. Interesting Properties of the Static Equilibrium

Effects of changes in X on welfare

3. Interesting Properties of the Static Equilibrium

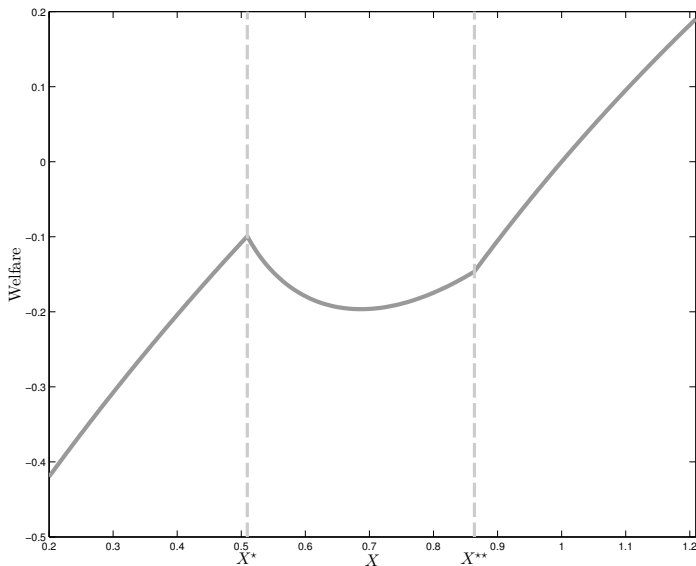
Effects of changes in X on welfare

Proposition 2 (Welfare)

- ▶ *If the economy is the unemployment regime and if τ is large enough (close enough to $\bar{\tau}$),*
- ▶ *then an increase in X leads to a fall in welfare.*

3. Interesting Properties of the Static Equilibrium

FIGURE 19 – Welfare as function of X



3. Interesting Properties of the Static Equilibrium

Introducing government spending

- ▶ Add a government in the morning
- ▶ $u(X + e + G_n) + \gamma \tilde{u}(G_u)$
- ▶ Government :
 - × purchase G_n that is perfectly substitutable with private consumption
 - × purchase G_w that is useless ($\gamma = 0$), or enters additively in utility
 - × Lump-sum taxes
 - × Balance budget

3. Interesting Properties of the Static Equilibrium

Introducing government spending (*continued*)

$$u(X + e + G_n) + \gamma \tilde{u}(G_u)$$

Proposition 3 (Fiscal Multitpliers)

- ▶ *An increase in G_n has no effect*
- ▶ *An increase in G_w increases activity.*
- ▶ *The multiplier de/dG_w is*
 - × *greater than one in the unemployment regime*
 - × *smaller than one in the full-employment regime*

3. Interesting Properties of the Static Equilibrium

Introducing government spending (*continued*)

Proposition 4 (Fiscal policy and welfare)

- ▶ *In the unemployment regime*
- ▶ *in the zone where a fall in X would increase welfare,*
- ▶ *an increase in G_w will increase welfare.*

0. Introduction

Roadmap

1. Model setup
2. Equilibrium
3. Interesting Properties of the Static Equilibrium
4. Extensions / Dynamics / Policy Trade-offs

4. Extensions / Dynamics / Policy Trade-offs

Relaxing functional-form assumptions

- ▶ Results are robust to :
 - × Relaxing functional assumptions (matching function)
 - × Other ways of splitting the surplus (Nash Bargaining, directed search)
 - × Introduction of productive capital
 - × Addition of another good in the morning (*cf* Krugman)
- ▶ Simple characterization is not always possible
- ▶ but main results hold.

4. Extensions / Dynamics / Policy Trade-offs

Endogenous imperfect insurance

- ▶ We endogenize the absence of unemployment insurance.
- ▶ Information friction : adverse selection.
- ▶ We can then compute the constrained efficient planner allocations

4. Extensions / Dynamics / Policy Trade-offs

Dynamic Setup

- ▶ An infinite number of periods t ,
- ▶ Each period consists of a morning and an afternoon
- ▶ The only financial trade is between morning and afternoon by assumption

▶

$$X_{t+1} = (1 - \delta)X_t + \gamma e_t$$

▶

$$U = \sum_{t=0}^{\infty} \beta^t \left(U(c_t) - \nu(\ell_t) + V(a_t) \right)$$

4. Extensions / Dynamics / Policy Trade-offs

Policy Trade-off

- ▶ When X is high, the economy will converge with the SS with inefficiently low demand on the way.
- ▶ Welfare today would be increased by stimulating demand today.
- ▶ But this would imply higher X tomorrow,
- ▶ And therefore lower consumption in all subsequent periods until the liquidation is complete.
- ▶ This tradeoff is aimed at capturing the tension between the Keynesian and Hayekian prescriptions in recession.

4. Extensions / Dynamics / Policy Trade-offs

Proposition 5 (Aggregate demand management is desirable)

- ▶ *Suppose the economy is in steady state in the unemployment regime.*
- ▶ *Then, to a first-order approximation, a (feasible) change in the path of expenditures from this steady state equilibrium will increase the present discounted value of expected welfare ...*
- ▶ *... if and only if it increases the presented discounted sum of the resulting expenditure path, $\sum_{i=0}^{\infty} \beta^i e_{t+i}$.*
- ▶ *Aggregate demand management is therefore desirable.*